

Week 1 Exercise Sheet - Reporting Fire Drill

Purpose

Students learn that an AI-generated report is not trustworthy until the metric definitions, source files, checks, and rejected agent assumptions are visible.

Materials

- Starter_Kit/sample_data/week1_customers.csv
- Starter_Kit/sample_data/week1_sales.csv
- Starter_Kit/sample_data/week1_support.csv
- Starter_Kit/sample_data/week1_metric_definitions.md
- Starter_Kit/templates/work_order_template.md
- Starter_Kit/templates/ai_use_log_template.md
- Starter_Kit/templates/defence_note_template.md

Exercise 1 - Metric Contract Clinic

Time: 15 minutes

Student task: In pairs, list the metrics the COO probably expects in the Monday KPI pack. For each metric, write the definition, source file, formula, and one possible ambiguity.

Suggested table:

Metric	Approved definition	Source	Formula	Possible ambiguity
Paid active accounts				
Net June revenue				
Open support tickets				

Instructor prompt: “If the agent used a different definition, how would you know before the COO sees the report?”

Expected evidence: A metric-definition table with at least one ambiguity clearly named.

Exercise 2 - Rewrite the Weak Prompt

Time: 15 minutes

Weak prompt:

Make me a KPI report from these CSVs.

Student task: Rewrite the prompt as an agent work order. It must include goal, inputs, acceptance criteria, and constraints.

Minimum requirements:

- Name all source files.
- State that the agent must not invent metric definitions.
- Require at least three smoke tests.
- Require a short explanation of any assumptions.

Instructor note: Push students to specify output format. A vague “make a report” prompt often produces a polished but uncheckable artefact.

Exercise 3 - Smoke-Test Hunt

Time: 25 minutes

Student task: Use the sample files to create at least three checks that would catch obvious errors.

Known checks:

- Paid active accounts should be 6.
- Recent-login customers should be 7.
- Gross June revenue should be 2860.
- Net June revenue should be 2660.
- Open support tickets should be 3.

Challenge: Write one check that would fail if the agent used recent-login customers instead of paid active accounts.

Expected evidence: A script, notebook, spreadsheet, or plain-language check list showing input, formula, expected value, and actual value.

Exercise 4 - Agent Failure Log

Time: 15 minutes

Student task: Ask the agent to draft the KPI workflow. Identify one suggestion you would reject or modify.

Log it using:

Agent suggestion	Risk	Evidence used	Team decision
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Instructor prompt: “The best log entry is not the longest. It is the entry that shows judgement.”

Exercise 5 - Defence Rehearsal

Time: 10 minutes

Each pair gives a 60-second defence:

1. What number do you trust?
2. What did the agent get wrong or almost get wrong?
3. Which check would catch the problem next time?

Listeners ask one question: “What would still make this report unsafe?”

Instructor Solution Notes

Strong teams usually separate paid active accounts from recent login customers. If a team reports only one “active” number, ask who owns the definition and whether a logged-in cancelled customer should count.

Common mistakes:

- Treating the sample metric definitions as optional reading.
- Reporting gross revenue when the stakeholder expects net revenue.
- Writing tests after seeing the answer without explaining what defect the test prevents.
- Logging “AI helped with code” instead of a specific accepted or rejected output.

Extension Options

- Add a generated executive summary that cites only checked numbers.
- Create a scheduled-run checklist.
- Add a “data freshness” check: no source file should be older than the reporting date.

Week 2 Exercise Sheet - The Board Number Is Wrong

Purpose

Students learn how to reconcile conflicting business numbers and how AI-generated joins can hide missing records.

Materials

- Starter_Kit/sample_data/week2_subscriptions.csv
- Starter_Kit/sample_data/week2_cancellations.csv
- Starter_Kit/sample_data/week2_product_events.csv
- Starter_Kit/templates/defence_note_template.md

Exercise 1 - Source Profile Before Joining

Time: 15 minutes

Student task: Profile each file before asking the agent to join anything.

Required profile:

Source file	Row count	Unique customer count	Date fields	Possible data-quality issue
subscriptions				
cancellations				
product events				

Instructor prompt: “A join without source counts is a trust fall.”

Exercise 2 - Join Choice Debate

Time: 20 minutes

Student task: Ask the agent for a join strategy. Before accepting it, compare three choices:

Join strategy	What it preserves	What it may hide	When it is appropriate
Inner join			
Left join from subscriptions			
Full outer review			

Expected discussion: The operational source of subscriptions should not lose customers simply because product usage is missing.

Exercise 3 - Quarantine Table

Time: 25 minutes

Student task: Build a quarantine table with records that should not be silently dropped.

Minimum fields:

- `customer_id`
- `source_seen_in`
- `issue_type`
- `why_it_matters`
- `recommended_action`

Starter defects to find:

- C204 is in subscriptions and cancellations but has no product event.
- C999 is in product events but not in subscriptions.
- C204 has a blank cancellation reason.

Exercise 4 - Validation Gate

Time: 20 minutes

Student task: Define a validation gate that should run before the board number is accepted.

Examples:

- Join coverage from subscriptions to product events must be at least a stated threshold.
- All cancelled customers must have billing-end dates.
- All quarantined rows must be reported, not deleted.

- The churn formula must name whether it uses request date or billing-end date.

Expected evidence: A test, SQL check, spreadsheet rule, or plain-language validation table.

Exercise 5 - Board Note

Time: 15 minutes

Write a short board note:

The number we recommend using is ____ because . *The previous numbers differed because* . Remaining limitations are ____.

Instructor Solution Notes

The important teaching point is not a single perfect churn number. It is the reconciliation logic. Strong teams can explain which rows were lost by an inner join and why a quarantined record is more honest than a silently cleaned table.

Common mistakes:

- Choosing the largest table as the source of truth without business justification.
- Removing unmatched rows to make the result clean.
- Treating missing cancellation reason as a harmless null.
- Failing to distinguish billing churn from product inactivity.

Extension Options

- Build an automated row-count bridge.
- Add a daily alert when join coverage changes.
- Write a short “definition owner” note naming Finance and Product responsibilities.

Week 3 Exercise Sheet - Limited Retention Budget

Purpose

Students learn that a model is useful only when it improves a constrained decision and survives leakage checks.

Materials

- Starter_Kit/sample_data/week3_retention_training.csv
- Starter_Kit/templates/defence_note_template.md
- Portfolio_and_Assessment_Model.md

Exercise 1 - Decision Frame

Time: 15 minutes

Student task: Complete the decision frame before building any model.

Question	Team answer
What action will the business take?	
Who is eligible?	
What is the capacity or budget limit?	
What is the cost of acting?	
What is the value of a successful save?	
What is the simplest baseline?	

Instructor prompt: “If no action changes, this is not a decision model yet.”

Exercise 2 - Leakage Hunt

Time: 20 minutes

Student task: Inspect every column and label it as allowed, target, leakage, or unclear.

Column	Allowed at decision time?	Reason
tenure_months		
monthly_fee		
usage_sessions_30d		
support_tickets_90d		
plan_type		
discount_last_90d		
post_offer_response		
retained_next_month		

Expected finding: `retained_next_month` is the target. `post_offer_response` is post-decision information and should not be used as a predictor.

Exercise 3 - Baseline Race

Time: 25 minutes

Student task: Build two simple baselines before accepting an agent-generated model.

Possible baselines:

- Highest support burden.
- Lowest product usage.
- Month-to-month plan with low usage.
- Highest monthly fee among low-usage customers.

For each baseline, record:

Baseline	Rule	Who is selected	Why it might work	Why it might fail
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Exercise 4 - Threshold Economics

Time: 25 minutes

Use these assumptions:

- Offer cost: 25 per customer.
- Expected saved margin: 120 for a truly saved customer.
- Capacity: contact only 30 percent of customers in this exercise.

Student task: Choose a threshold or top-N policy. Explain why it is better than “send to everyone” or “use 0.5 as the default threshold.”

Expected-value structure:

Expected value = saved customers * 120 - contacted customers * 25

Exercise 5 - Decision Memo Clinic

Time: 15 minutes

Draft a one-page memo with this structure:

1. Recommended decision policy.
2. Baseline comparison.
3. Leakage check.
4. Cost or capacity rationale.
5. What should not be claimed yet.

Instructor Solution Notes

Strong teams do not worship model metrics. They explain the action, capacity, and threshold. They can say which columns were excluded and why.

Common mistakes:

- Using `post_offer_response` because it improves the score.
- Reporting accuracy without comparing to the majority class.
- Letting the agent pick a threshold without business logic.
- Writing a memo that says “the model predicts churn” without saying what action follows.

Extension Options

- Add fairness or service-quality slices.
- Test whether the policy over-targets high-fee customers.
- Compare model recommendation with a transparent business rule.

Week 4 Exercise Sheet - From Notebook to Tool

Purpose

Students learn how to turn an analysis into an internal analytics product with explicit inputs, outputs, error handling, tests, and operating notes.

Materials

- Week 3 decision rule or model output.
- Week 4 request schema in `Sample_Data_and_Scenarios.md`.
- `Starter_Kit/templates/runbook_template.md`
- `Starter_Kit/templates/peer_demo_review_template.md`

Exercise 1 - Product Boundary

Time: 15 minutes

Student task: Define what the tool does and does not do.

Boundary question	Team answer
Who is the user?	

Boundary question	Team answer
What input do they provide?	
What output do they receive?	
What should the tool refuse?	
What is outside scope?	

Instructor prompt: “A tool boundary protects both the user and the analytics team.”

Exercise 2 - Input Contract

Time: 20 minutes

Use this suggested request:

```
{
  "customer_id": "C301",
  "tenure_months": 14,
  "monthly_fee": 79,
  "usage_sessions_30d": 5,
  "support_tickets_90d": 3,
  "plan_type": "monthly"
}
```

Student task: Define required fields, types, allowed values, and rejection rules.

Minimum rejection rules:

- Missing `customer_id`.
- Negative `tenure_months`.
- Unknown `plan_type`.
- Missing usage or support data.

Exercise 3 - Bad-Input Lab

Time: 25 minutes

Student task: Create three invalid requests and decide the expected response before implementing anything.

Bad request	Expected response	Why this matters
Missing customer ID		

Bad request	Expected response	Why this matters
Negative tenure		
Unknown plan type		

Then ask the agent to add the validation logic and tests.

Exercise 4 - Handoff Test

Time: 20 minutes

Swap tools with another team. The receiving team must run the tool using only the README, runbook, or instructions provided.

Reviewer questions:

- Could you start it without asking the original team?
- Did the good request work?
- Did the bad request fail clearly?
- Was the output understandable?

Exercise 5 - Runbook Drill

Time: 15 minutes

Complete the runbook sections:

- Purpose.
- Owner.
- Start procedure.
- Health checks.
- Known failure modes.
- Rollback procedure.

Instructor Solution Notes

Strong teams make failure boring. Bad input should produce a controlled rejection, not a crash or a confident nonsense answer.

Common mistakes:

- Shipping a notebook with hidden state.
- Accepting any JSON because the happy path works.
- Writing “contact IT” as the only recovery step.
- Failing to record which model, rule, or data version the tool used.

Extension Options

- Add request logging.
- Add a simple latency or cost estimate.
- Add a version field to every output.

Week 5 Exercise Sheet - The Executive Question

Purpose

Students learn to evaluate an analyst agent that answers business questions with evidence, uncertainty, and refusal behaviour.

Materials

- Starter_Kit/sample_data/week5_knowledge_base.md
- Starter_Kit/sample_data/week5_eval_cases.csv
- Starter_Kit/templates/ai_use_log_template.md
- Starter_Kit/templates/defence_note_template.md

Exercise 1 - Answer Anatomy

Time: 15 minutes

Student task: Break a safe executive answer into five parts:

1. Observed pattern.
2. Evidence.
3. Plausible explanation.
4. Uncertainty or limitation.
5. Recommended next test.

Apply this structure to the question: “Why did churn rise in June?”

Exercise 2 - Evaluation Set Expansion

Time: 20 minutes

Start from `week5_eval_cases.csv`. Add three new cases:

- One factual question.
- One unsupported causal question.

- One out-of-scope policy question.

For each case, define expected behaviour:

Question	Should answer?	Expected evidence	Should refuse?	Why

Exercise 3 - Citation Support Check

Time: 25 minutes

Student task: Ask the agent to answer two evaluation questions. For every sentence with a citation, check whether the cited source supports the exact claim.

Use this marking:

- **Supported:** the source directly supports the claim.
- **Partially supported:** the source supports part of the claim but the answer overstates it.
- **Unsupported:** the cited source does not support the claim.

Instructor prompt: “A citation is not decoration. It is a promise that the evidence says what the answer says.”

Exercise 4 - Refusal Rehearsal

Time: 15 minutes

Students rewrite unsafe answers into safe refusals.

Unsafe answer:

The May price increase caused churn to rise in June.

Safer answer:

The evidence does not support that causal claim. The pricing note says the May price increase applied to new monthly-plan customers, while existing customers were not moved to the new price in June. A safer next step is to compare churn among new customers exposed to the new price with similar customers not exposed.

Exercise 5 - CEO-Ready Answer

Time: 15 minutes

Write a final answer that could be sent to the CEO. It must include:

- One observed pattern.
- Two cited sources.
- One uncertainty statement.
- One recommended next test.
- One claim the team refuses to make.

Instructor Solution Notes

Strong teams separate observation from explanation. They do not claim causality when the evidence only supports timing, association, or a plausible hypothesis.

Common mistakes:

- Treating a source mention as support for any related sentence.
- Evaluating only easy questions.
- Refusing too vaguely: “I cannot answer” without saying what evidence is missing.
- Letting the agent write an executive answer that the student cannot defend.

Extension Options

- Compare two prompts on the same evaluation set.
- Add an answer quality rubric with factuality, citation support, uncertainty, and refusal.
- Log prompt, retrieved source, answer, and review decision for each run.

Week 6 Exercise Sheet - Launch Review

Purpose

Students learn to defend a launch decision using evidence from the full portfolio. The final session tests operational judgement, not just presentation polish.

Materials

- Weekly artefacts from Weeks 1-5.
- Starter_Kit/templates/evidence_map_template.md
- Portfolio_and_Assessment_Model.md
- Sample_Data_and_Scenarios.md Week 6 incident injects.

Exercise 1 - Evidence Map

Time: 25 minutes

Student task: Map each launch claim to evidence.

Launch claim	Evidence	What it proves	What it does not prove	Owner
KPI pack can be rerun	Churn number is reconciled	Decision model avoids leakage	Tool rejects bad input	Analyst answer is source-grounded

Instructor prompt: “A claim without an evidence link is a hope, not a launch argument.”

Exercise 2 - Launch Checklist

Time: 20 minutes

Students complete a launch checklist:

- Scope of use.
- No-ship conditions.
- Required evidence.
- Monitoring owner.

- Support owner.
- Rollback path.
- Known limitations.

Exercise 3 - Incident Drill

Time: 30 minutes

The instructor gives one inject during each team's demo.

Inject examples:

- The active customer count is outside the expected range.
- A user sends a request with missing fields.
- A stakeholder asks the team to claim the campaign caused churn reduction.
- The latest data-quality gate failed.
- No one knows who monitors the next run.

Student response must include:

1. What pauses or stops.
2. Which evidence or check is used.
3. Who owns the next action.
4. What is communicated to stakeholders.

Exercise 4 - Viva Circle

Time: 20 minutes

Students answer individual questions in rotating pairs:

- Show one place where the agent was wrong.
- Show evidence that changed your mind.
- What would fail at ten times the volume?
- Which portfolio claim is weakest?
- What should not be shipped yet?

Peer listener records one follow-up question.

Exercise 5 - Final Recommendation

Time: 15 minutes

Students write the final recommendation:

We recommend ship / limited pilot / no ship because . *The strongest evidence is* . The main residual risk is . *The next owner is* .

Instructor Solution Notes

Strong final demos are specific about scope. A team can receive a high mark with a “limited pilot” or “no ship yet” recommendation if the evidence and ownership logic are strong.

Common mistakes:

- Turning the final demo into a feature tour.
- Hiding weaknesses instead of owning residual risk.
- Making a broad launch claim from narrow evidence.
- Saying “the agent will monitor it” without naming a human owner.

Extension Options

- Run a cross-team red-team review.
- Record a two-minute executive launch recommendation.
- Add a post-launch monitoring dashboard design.